

COSINE RULE

Higher Tier

Direct calculations, proof and multi-step problems

Name	
Date	

11 questions	49 marks	2017-2025
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Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

Using this booklet

Questions are arranged by exam series. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

This is an independently compiled practice resource, not an official Pearson examination paper. The source collection covers 2017-2025. Questions are included only in the booklet that matches their main purpose.

Question index

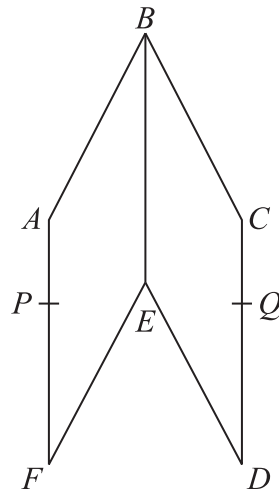
Higher Tier | 11 questions | 49 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	June 2017	1MA1/1H	Q22	5	3
02	November 2017	1MA1/3H	Q17	5	4
03	June 2018	1MA1/3H	Q17	5	5
04	June 2019	1MA1/3H	Q23	5	6
05	November 2019	1MA1/3H	Q18	5	7
06	November 2021	1MA1/2H	Q15(a)	3	8
07	June 2022	1MA1/2H	Q18	4	9
08	June 2023	1MA1/3H	Q14	4	10
09	November 2023	1MA1/1H	Q20	4	11
10	November 2024	1MA1/3H	Q17	4	12
11	June 2025	1MA1/2H	Q23	5	13

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

22 The diagram shows a hexagon $ABCDEF$.



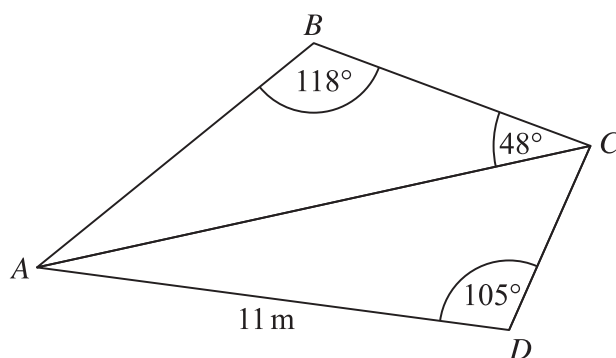
$ABEF$ and $CBED$ are congruent parallelograms where $AB = BC = x$ cm.
 P is the point on AF and Q is the point on CD such that $BP = BQ = 10$ cm.

Given that angle $ABC = 30^\circ$,

prove that $\cos PBQ = 1 - \frac{(2 - \sqrt{3})}{200}x^2$

(Total for Question 22 is 5 marks)

17 ABC and ADC are triangles.



The area of triangle ADC is 56 m^2

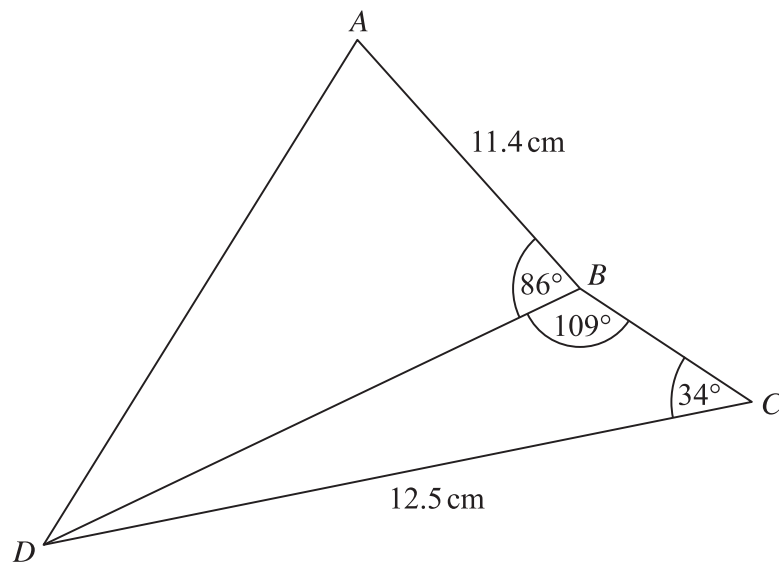
Work out the length of AB .

Give your answer correct to 1 decimal place.

..... m

(Total for Question 17 is 5 marks)

17



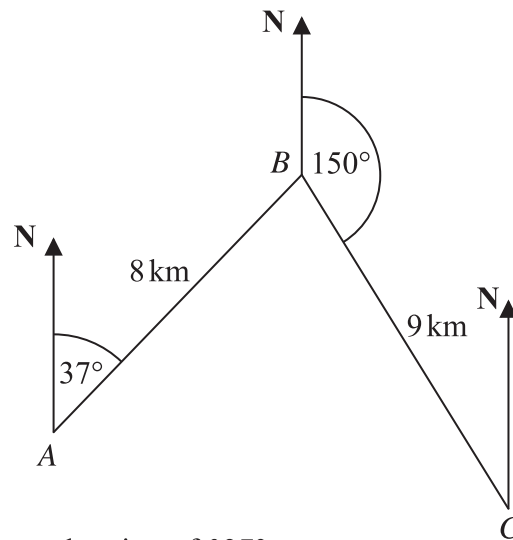
Work out the length of AD .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 17 is 5 marks)

23 The diagram shows the positions of three towns, Acton (A), Barston (B) and Chorlton (C).

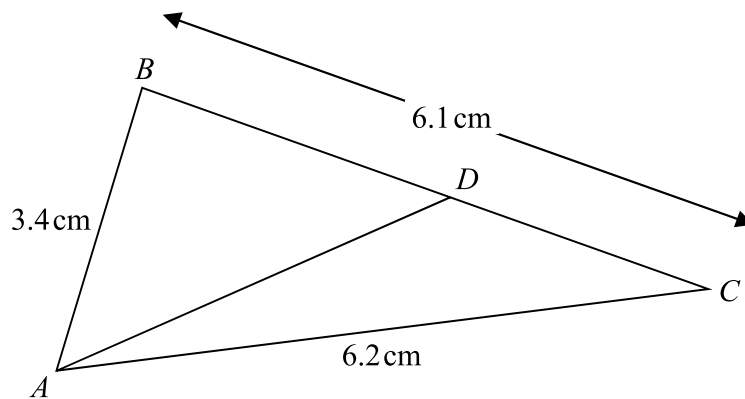


Barston is 8 km from Acton on a bearing of 037°
Chorlton is 9 km from Barston on a bearing of 150°

Find the bearing of Chorlton from Acton.
Give your answer correct to 1 decimal place.
You must show all your working.

(Total for Question 23 is 5 marks)

18 The diagram shows triangle ABC .



$$AB = 3.4\text{ cm} \quad AC = 6.2\text{ cm} \quad BC = 6.1\text{ cm}$$

D is the point on BC such that

$$\text{size of angle } DAC = \frac{2}{5} \times \text{size of angle } BCA$$

Calculate the length DC .

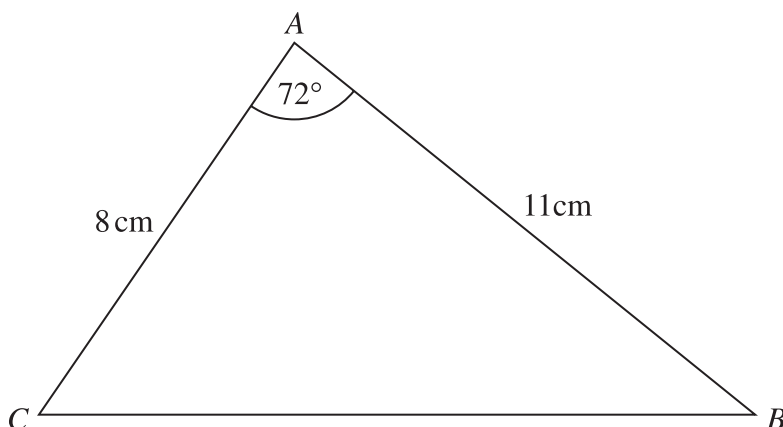
Give your answer correct to 3 significant figures.

You must show all your working.

..... cm

(Total for Question 18 is 5 marks)

15 Here is triangle ABC .

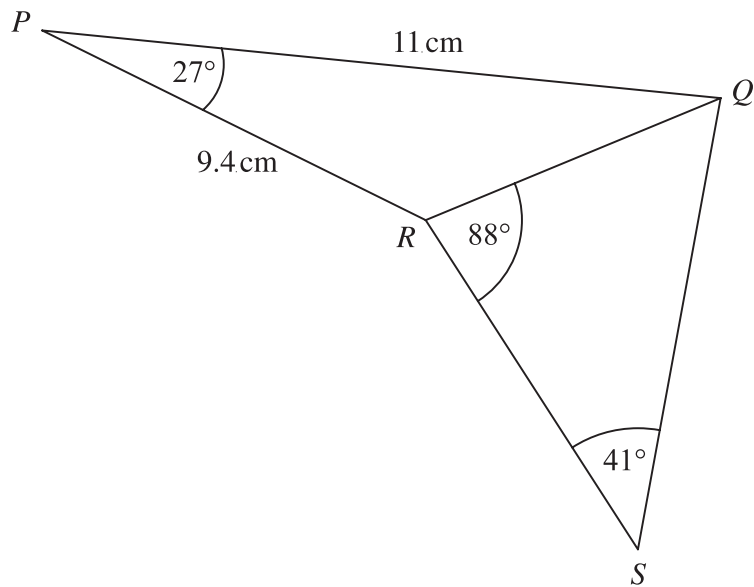


- (a) Find the length of BC .
Give your answer correct to 3 significant figures.

..... cm
(3)

Additional working space

18 PQR and QRS are triangles.



Calculate the length of QS .

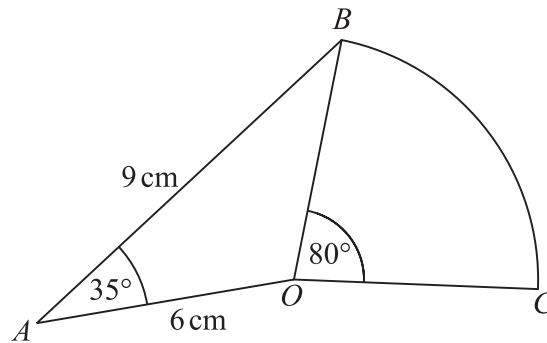
Give your answer correct to 3 significant figures.

You must show all your working.

..... cm

(Total for Question 18 is 4 marks)

- 14 OAB is a triangle.
 OBC is a sector of a circle, centre O .

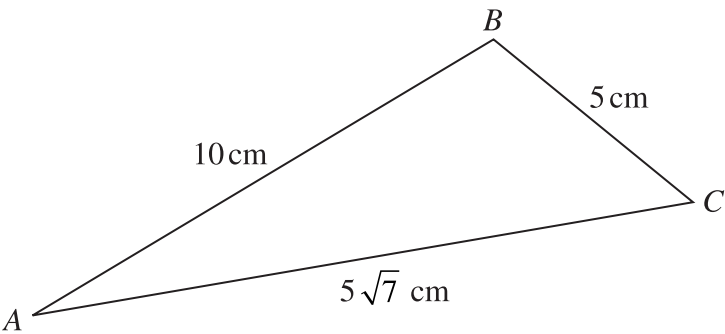


Calculate the area of OBC .
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 14 is 4 marks)

20 Here is triangle ABC .



Find the size of angle ABC .
You must show all your working.

(Total for Question 20 is 4 marks)

Additional working space

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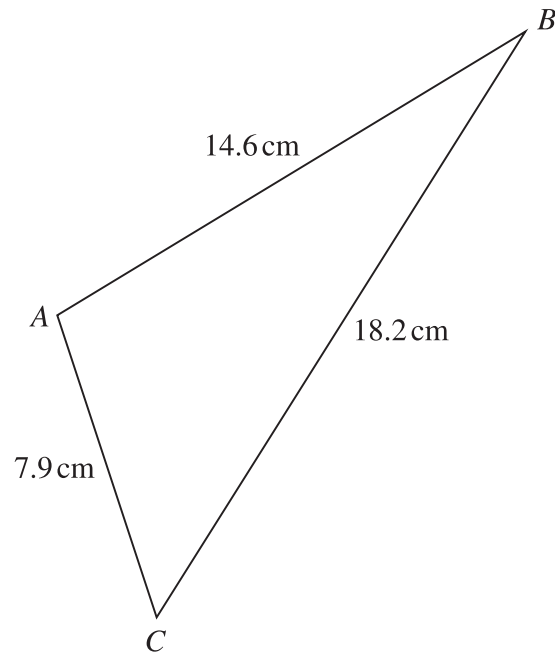
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17 Here is triangle ABC .

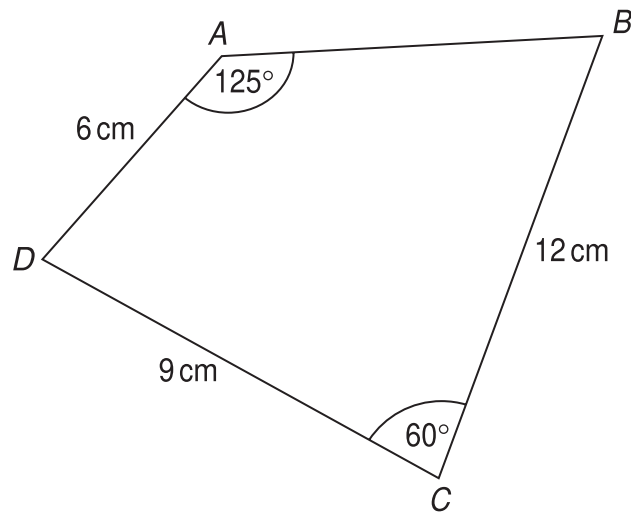


Work out the area of triangle ABC .
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 17 is 4 marks)

23 $ABCD$ is a quadrilateral.



Find the size of angle ABC .

Give your answer correct to the nearest degree.

.....
(Total for Question 23 is 5 marks)
